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Systems Thinking and Systems Engineering

PRASANNA RAMAMURTHY
TECHNICAL FELLOW– SYSTEMS ENGINEERING

17th OCTOBER 2026

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PRASANNA RAMAMURTHY

TECHNCIAL FELLOW – SYSTEMS ENGINEERING, AVIONICS



EDUCATION AND CAREER HISTORY

University of Madras

Bachelor of Engineering
(Electronics & Communication)
2001

Master of Engineering
(Avionics)
2003

PhD
UAS Traffic Management
(in progress since 2022)

2003
Systems Engineer

2011
Senior Systems Engineer

2016
Technical Fellow– Systems Engineering

Present

PROFESSIONAL CAREER

- Systems engineering expert with specialization in Avionics and Military mission systems and industry experience of 23 years.
- Leading the development of technologies and solutions in the areas of Advanced Air Mobility, Air Traffic Management, Wireless Sensor Networks and Integrated Vehicle Health Management (IVHM) system at Collins.
- Leading initiatives for technical capability development and systems engineering discipline maturity at Collins India.
- Made significant contributions to DRDO's MiG-27 and Su-30 MI avionics upgrade programs
- Systems Lead for Avionics, Radar and Simulation systems at Airbus Defence & Space

PATENTS, PUBLICATIONS, AWARDS

- Performance excellence award from DRDO for MiG-27 Avionics upgrade program
- 8 US patents, 1 trade secret, 7 publications in international conferences and journals
- 2 Partner awards for WAIC technology development at Collins Aerospace
- Collins India Innovation Award 2019

PARTICIPATION / LEADERSHIP IN INDUSTRY COMMITTEES

- President, INCOSE India Chapter
- Member, SAE HM-1 committee & ARP 6887 Work group
- Co-Lead, UTM Standards work group, Bureau of Indian Standards
- Reviewer & Track chair, INCOSE & SAE conferences

Bureau of Indian Standards (BIS)

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Outline

1. What's a System?
2. Systems Thinking – An introduction
3. Systems Thinking Tools
4. Context of Systems Engineering
5. Systems Engineering – An introduction
6. Whats' new in Systems Engineering

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System

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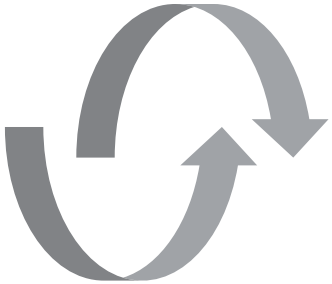
What is a system?

- A group of interacting, interrelated or interdependent parts that form a complex and unified whole that has a specific purpose.

- Interrelationship and Interdependence are the key; without them they are just **collection** of elements and not a **system**.



Collection



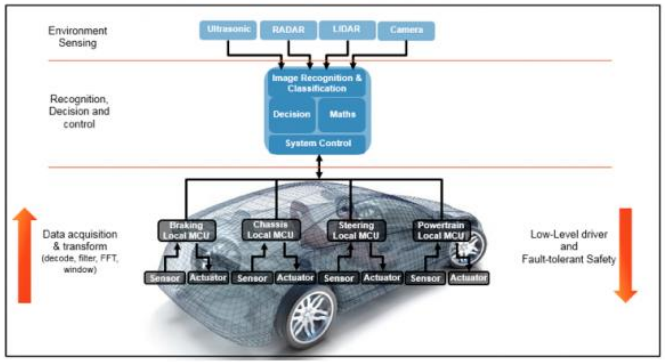


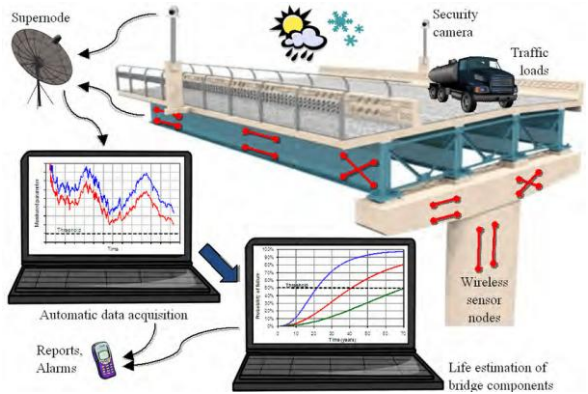
System

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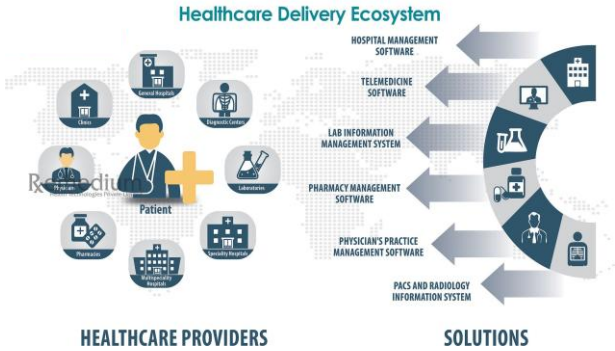
What is a system?









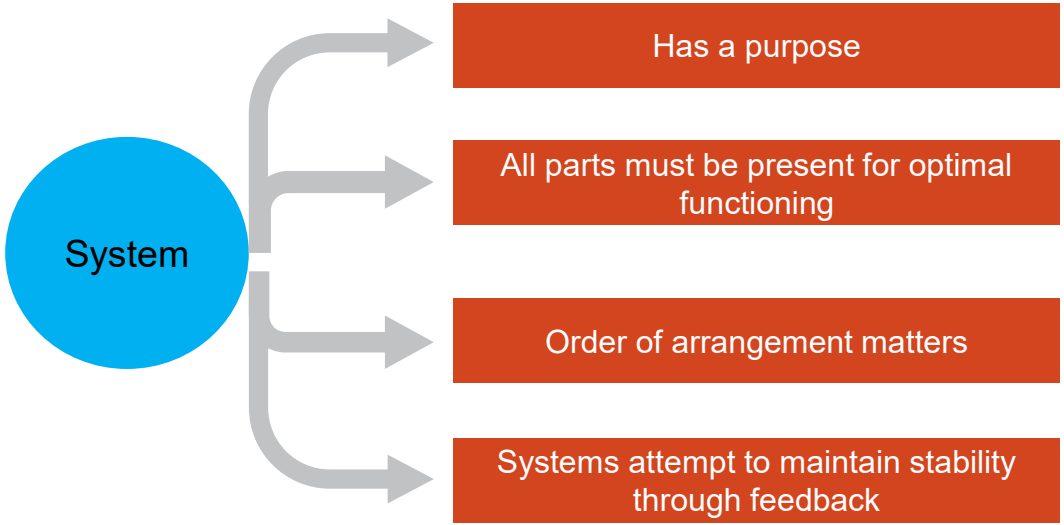


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Characteristics of a system

A combination of interacting elements organized to achieve one or more stated purposes (ISO/IEC/IEEE 15288)

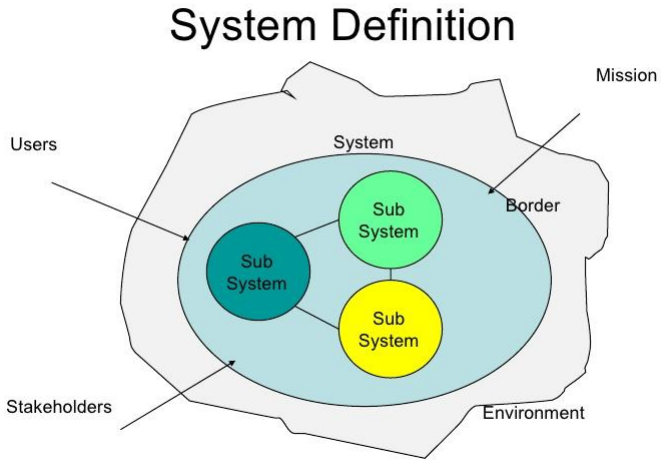
An integrated set of elements, subsystems, or assemblies that accomplish a defined objective. These elements include products (hardware, software, firmware), processes, people, information, techniques, facilities, services, and other support elements. (INCOSE)

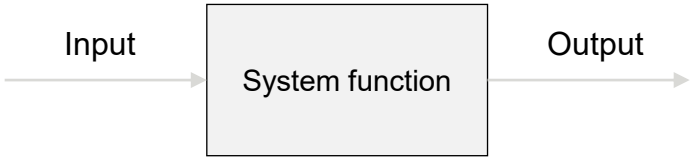


Source: Introduction to Systems thinking, Daniel H.Kim

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Characteristics of a system





Physical

- BOUNDARY
- ENVIRONMENT
- ELEMENTS
- INTERFACES
- ARRANGEMENT OF ELEMENTS

Operational / Functional

- INPUT
- FUNCTION / PROCESS
- OUTPUT

Quality

- FUNCTIONALITY
- RELIABILITY
- MAINTAINABILITY
- AVAILABILITY
- ……

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An Engineered System

- Are created, used and sustained to achieve a purpose, goal or mission that is of interest to an enterprise, team, or an individual.
- Require a commitment of resources for development and support.
- Are driven by stakeholders with multiple views on the use or creation of the system, or with some other stake in the system, its properties or existence.
- Contain engineered hardware, software, people, services, or a combination of these.
- Exist within an environment that impacts the characteristics, use, sustainment and creation of the system.

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Engineered Systems







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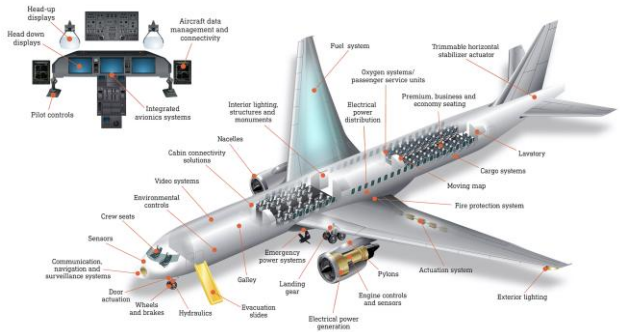
Safety Critical Systems

- **Safety critical systems** - Systems whose operation may have an impact on the human safety and life
  - Transportation systems
  - Aircrafts
  - Nuclear reactors
  - Medical surgical equipment
- **Criticality level** – Proportional to the severity of the impact and the number of people that are impacted
- Imperative to incorporate safety features right from the product conception stage
- Commercial aircrafts are highly safety critical system due to the volume of the people that they serve









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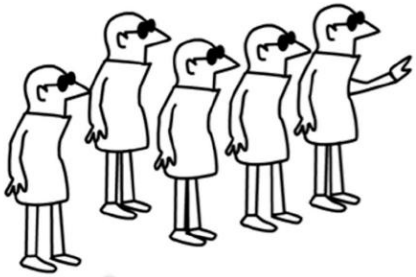
Systems Thinking

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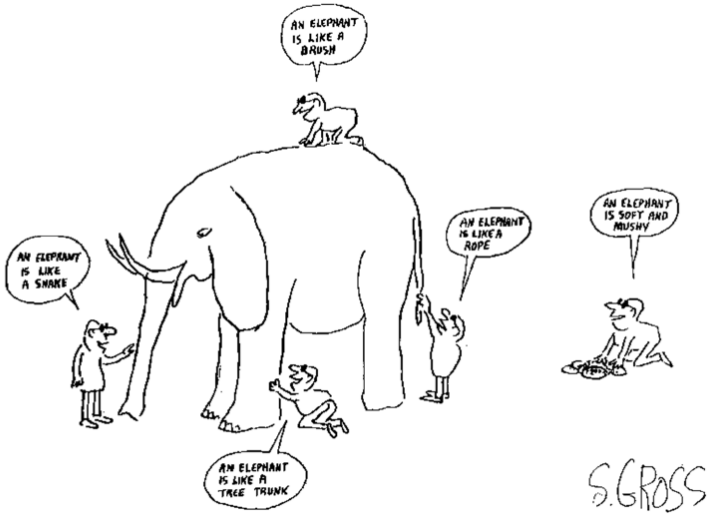
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Systems Thinking

- Seeing things only in parts leads to ambiguity
- See the parts as part of the whole





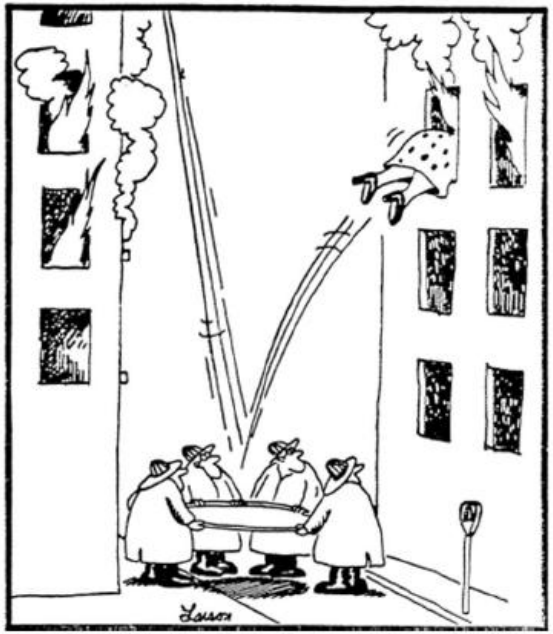


Source: Introduction to Systems Thinking Principles and Analytical Tools, https://www.unescap.org/

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Systems Thinking

- Partial solutions to a problem might lead to new problems; think holistically



Artist:
Gary Larson

Source: Introduction to Systems Thinking Principles and Analytical Tools, https://www.unescap.org/

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Systems Thinking

- A way of making sense of the complexity of the world by looking at it in terms of **wholes and relationships rather than by splitting** it down into its parts.
- A vantage point from which you see a whole, a web of relationships, rather than focusing only on the detail of any particular piece. Events are seen in the larger context of a pattern that is unfolding over time.
- An approach that is based on the belief that **the component parts of a system will act differently when isolated from the system's environment or other parts of the system**.

Systems thinking is the practice of examining and improving our mental models to understand the world around us (objects, groups, societies, relationship, etc) better and solve the complex problems associated with it.

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Systems Thinking

Key Components

Multiple Perspectives

- Who are the key stakeholders?
- What stakes do they have?
- What are the different ways in which the situation can be framed or understood by the stakeholders?

Interconnections

- How do the elements (components, stakeholders, knowledge, etc.) interconnect?
- What patterns emerge from these relationships, with what consequences, and for whom?

Boundaries

- Define scope and scale
- Are other boundaries possible and feasible?
- What constitutes an improvement –and how this might be different for different stakeholders?

Influences

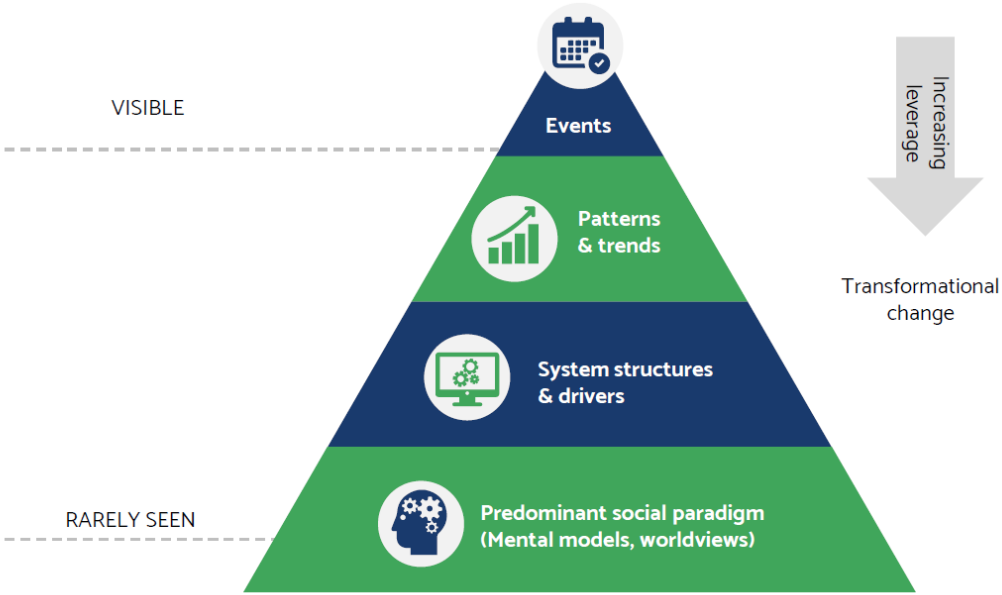
- What drives the systems in particular directions?
- Identify …. drivers, trends, enablers, blocks, leverage points
- Leverage points are seen as key points with which to intervene in complex systems

Source: Will Allen & Margaret Kilvington, An introduction to systems thinking and systemic design –concepts and tools, https://learningforsustainability.net/

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Systems Thinking

Iceberg model



- What is the fastest way to react to this event NOW?
- What trends and/or patterns are occurring?
- What mental/organizational structures create the patterns?
- What are the stated/unstated visions that generate this structure?

Source: Will Allen & Margaret Kilvington, An introduction to systems thinking and systemic design –concepts and tools, https://learningforsustainability.net/

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Systems Thinking

A Language and tool

To enhance our understanding and communicate on the system problems effectively, we need a language naturally suited to the task.

- Focuses on “interdependencies”
- Embodies a worldview that looks at the “whole” rather than “parts”
- Offers a “visual” language
- Allows “examination” and “enquiry”
- Adds “precision”
- Forces an “explicitness” of mental models

Source: Systems Thinking Tools, Pegasus Communications Inc. 781.398.9700

This document does not include any export controlled technical data.

Systems Thinking

Tools

- Behavior over time graph
- Causal loop diagram
- Systems archetype



dynamic thinking tools

- Mind maps
- Double Q diagram / Fishbone diagram



brainstorming tools

Systems Thinking Tools



structural thinking tools

- Graphical function diagram
- System Interaction Diagram
- Structure-behavior pair
- Policy structure diagram



computer-based tools

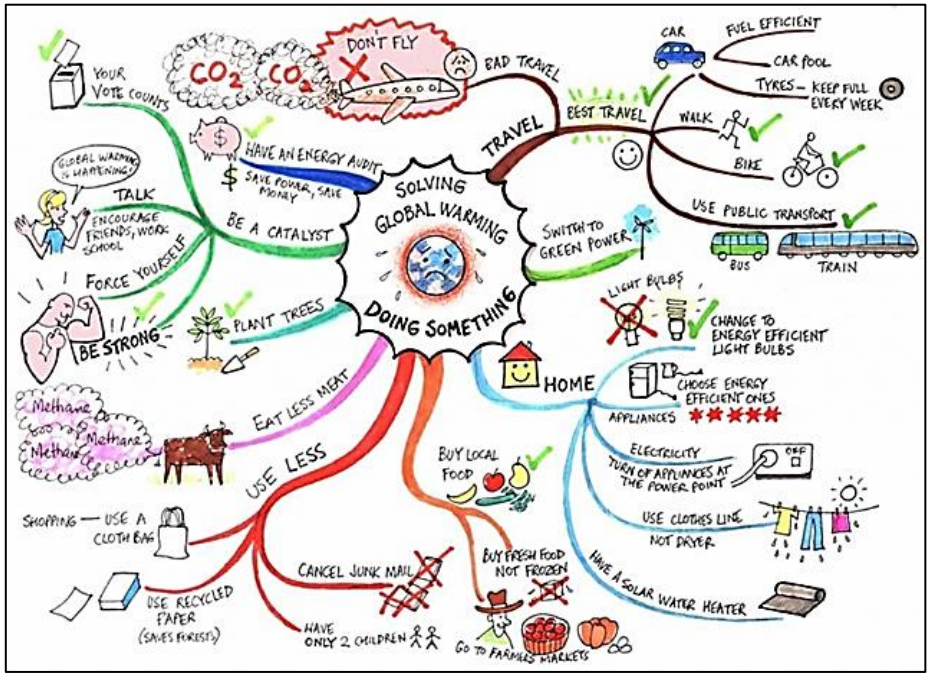
- Computer Model
- Management Flight Simulator
- Learning Laboratory

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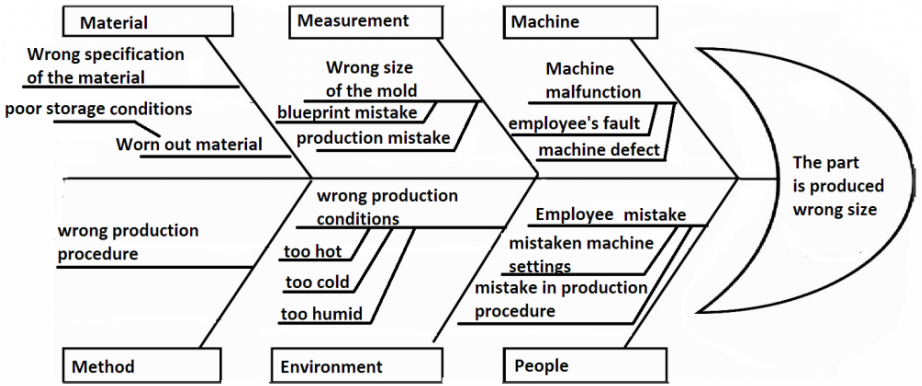
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Systems Thinking

Brainstorming Tools – Mind Map



Brainstorming Tools – Fishbone Diagram

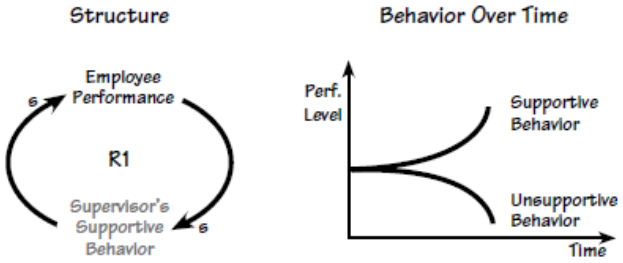


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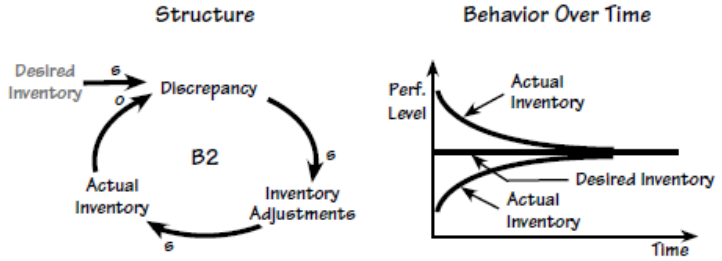
Systems Thinking

Dynamic Thinking Tools – Causal loop and Behavior over time (BoT) diagrams

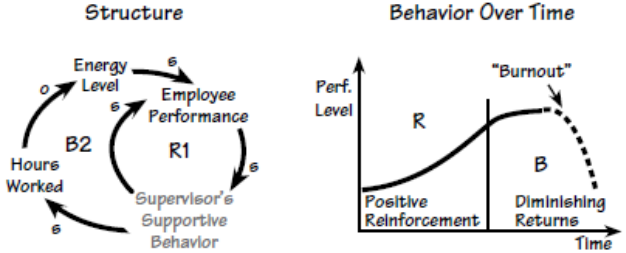
Reinforcing Loop



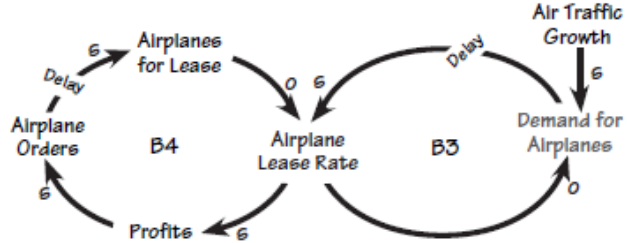
Balancing Loop



Combination of Balancing and Reinforcing Loops



Balancing Loop with Delays



Source: Systems Thinking Tools, Pegasus Communications Inc. 781.398.9700

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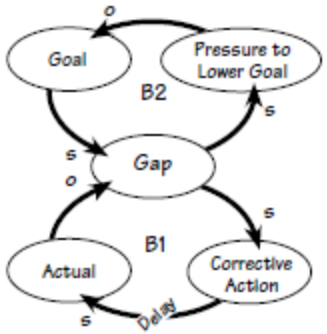
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Systems Thinking

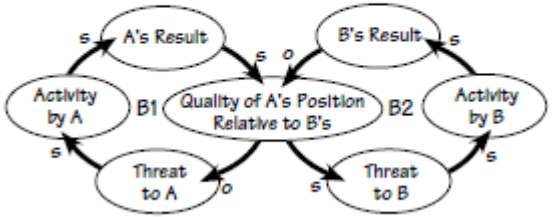
Dynamic Thinking Tools – System Archetypes

Predefined set of models representing certain common types of dynamics observed in systems. Users can fit their problems to one of these to analyze them.

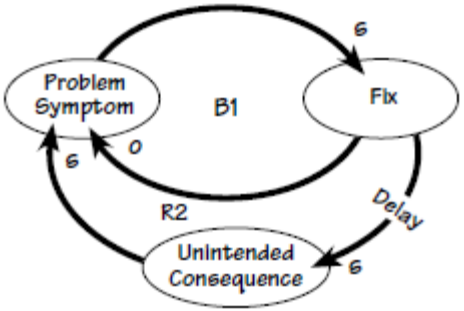
Drifting Goals



Escalation



Fixes That Fail



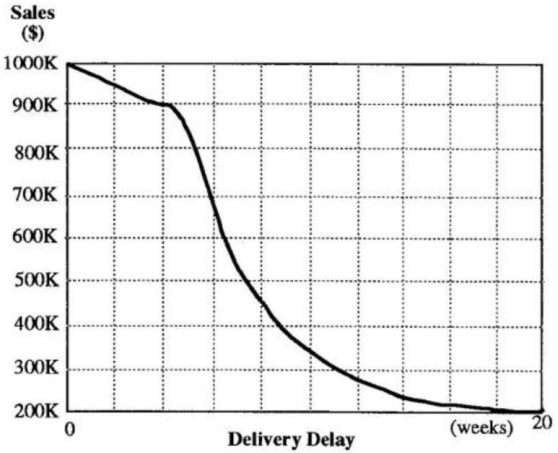
Source: Systems Thinking Tools, Pegasus Communications Inc. 781.398.9700

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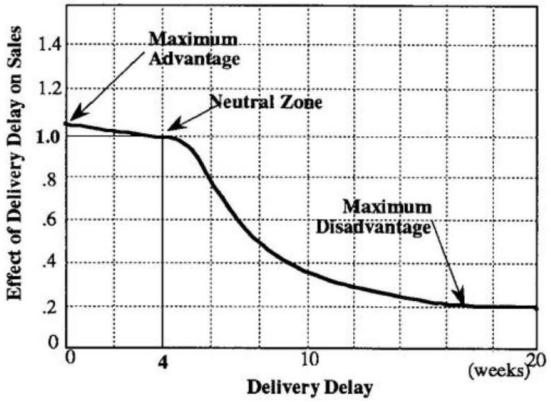
Systems Thinking

Structural Thinking Tools – Graphical Function Diagrams

- Provides the causal relationship between two variables.
- Represents variation of one variable w.r.t another.
- The relationship between variables established through system understanding rather than historical data (as in the case of a typical scatter diagram)
- Questions to answer before creating GFD
  - What is the causal relationship between the variables based on system understanding?
  - Are there any “neutral zones” where the variable on the y-axis is not affected by the variable in x-axis?
  - What are the extreme values that both variables can assume?



Source: DANIEL KIM, GRAPHICAL FUNCTIONS: “SEEING” THE FULL STORY, HTTPS://THESYSTEMSTHINKER.COM/

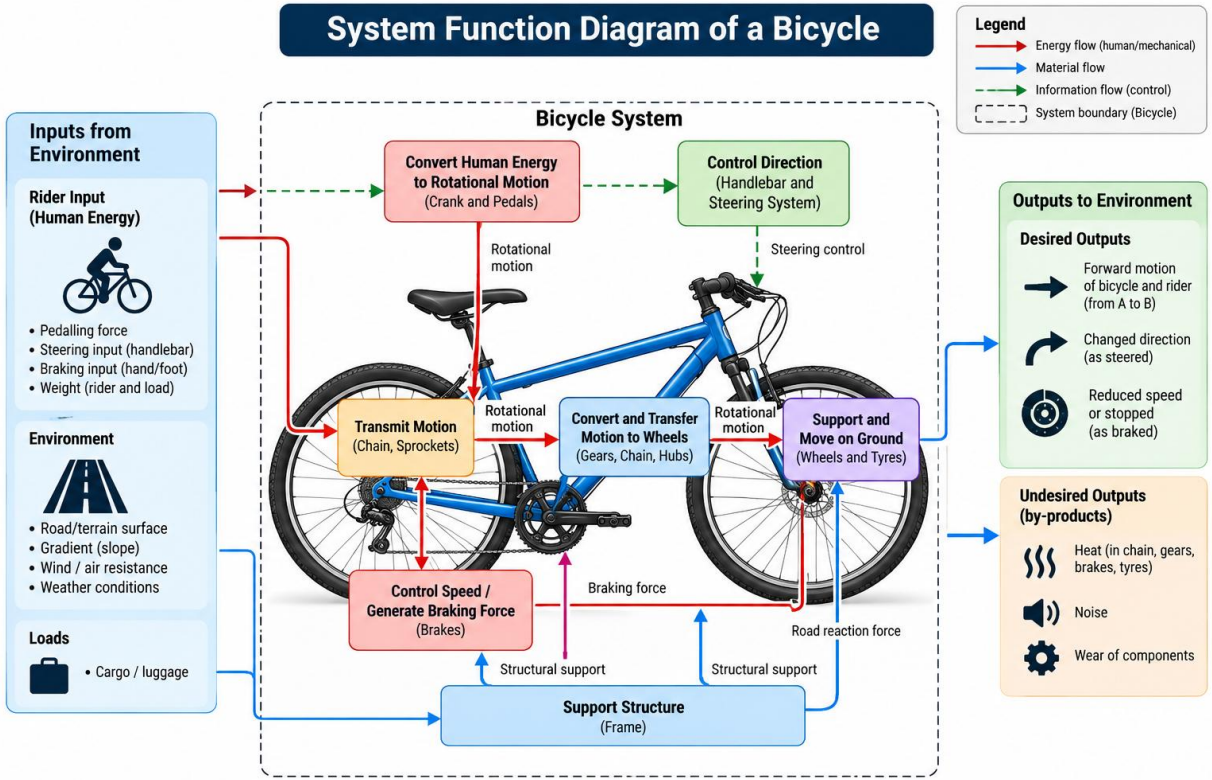


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Systems Thinking

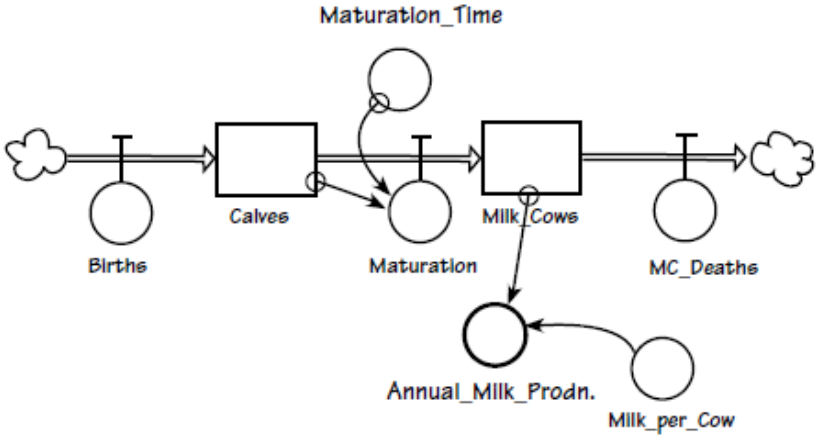
Structural Thinking Tools – System Function Diagrams

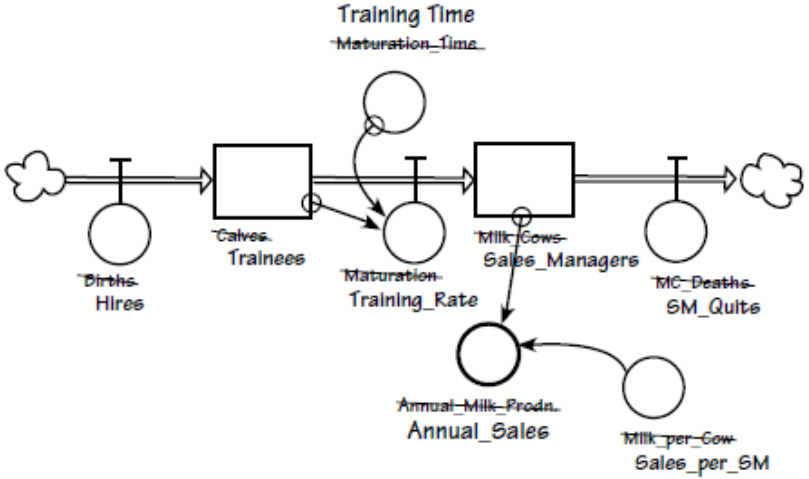


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Systems Thinking

Structural Thinking Tools – Accumulators and Flows





Source: Daniel H.Kim, Systems Thinking Tools, Pegasus communications Inc.Waltham, 1994

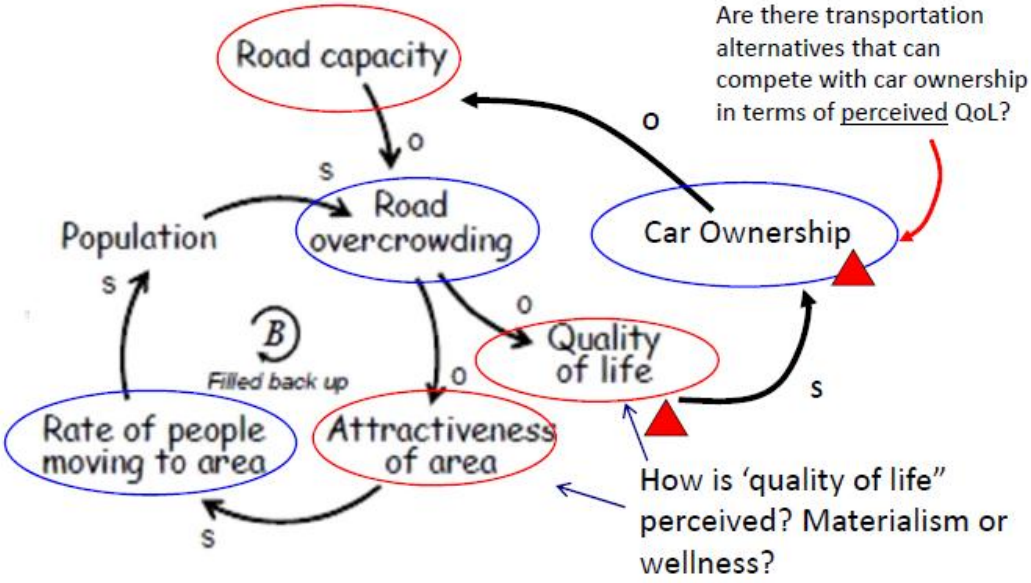
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Systems Thinking

Intervention Points

To find the most important places for intervention to change the long-term behavior of a system.



Source: Senge, Peter, The Fifth Discipline, 1996.

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Systems Thinking

Intervention Points

- ❑ Numbers
- ❑ Buffers
- ❑ Stock and flow structures
- ❑ Delays
- ❑ Balancing feedback loops
- ❑ Reinforcing feedback loops

- ❑ Information flows
- ❑ Rules
- ❑ Self-organization
- ❑ Goals
- ❑ Paradigms
- ❑ Transcending Paradigms



Source: Will Allen & Margaret Kilvington, An introduction to systems thinking and systemic design –concepts and tools, https://learningforsustainability.net/

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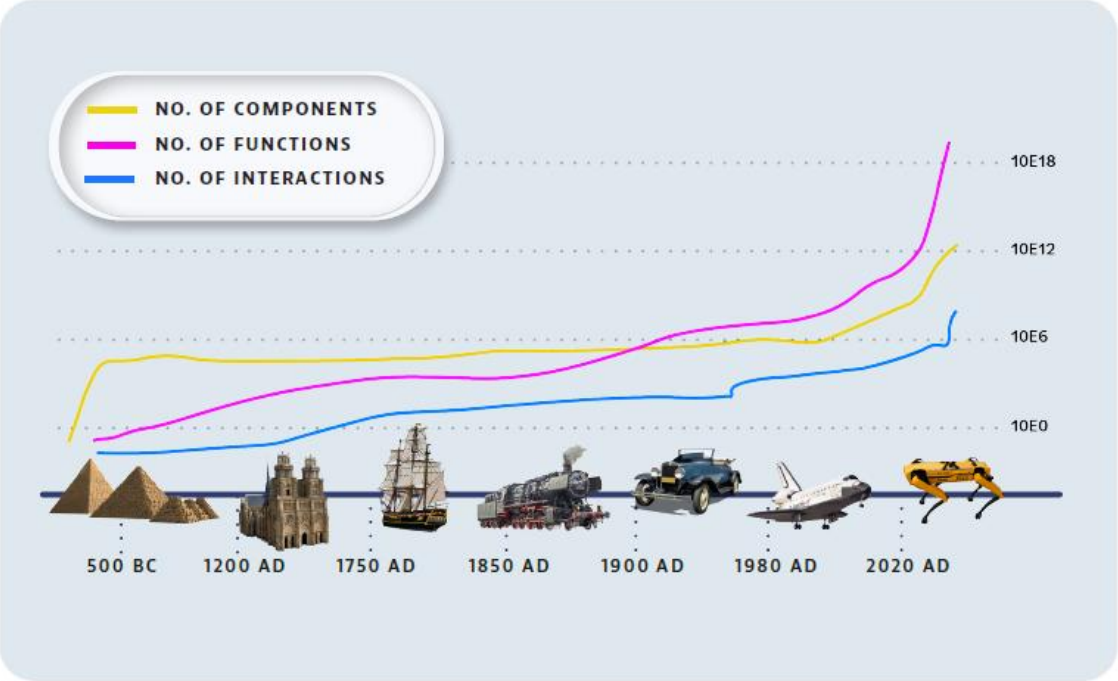
System Complexity

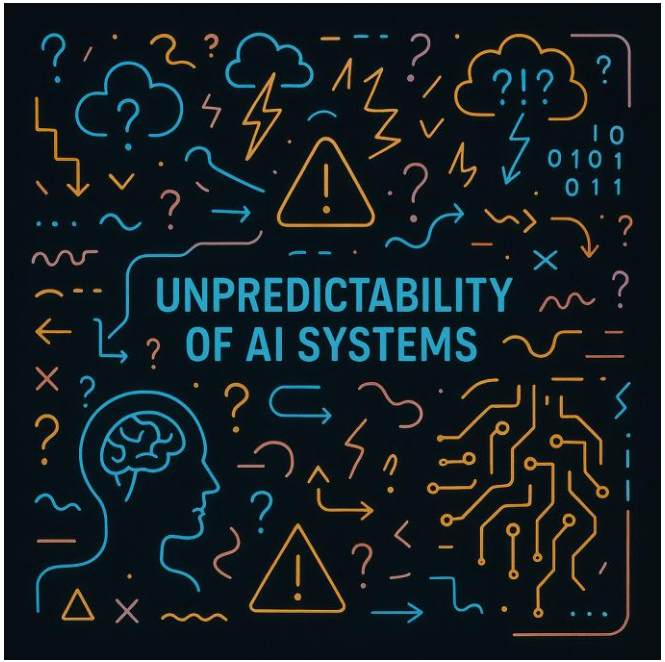
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System Complexity

Difficulty in understanding, analyzing and predicting the system behavior



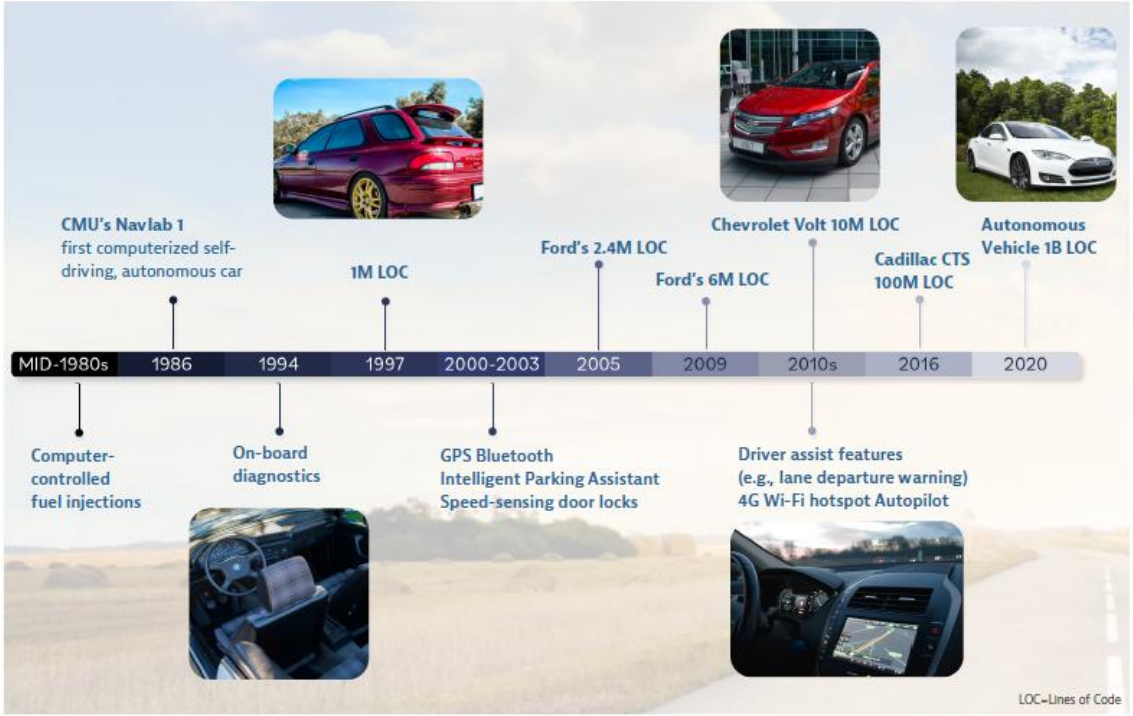


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System Complexity

Difficulty in understanding, analyzing and predicting the system behavior



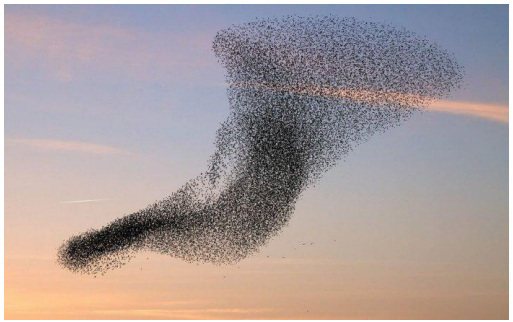
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System Complexity

Emergence

System-level properties, behaviors or capabilities that arise from interactions among components and their environment, rather than being attributable to any one component in isolation.



Source: https://moya-planeta.ru/travel/view/pochemu_skvortsy_tantsuyut_47136

- Positive / Negative
- Difficult to predict in complex systems
- In AI Systems...?



Source: https://thecityfix.com/blog/from-moving-cars-to-moving-people-scaling-up-safer-streets-in-indian-cities/



Source: https://www.aicerts.ai/news/chinas-ai-drone-swarm-test-redefines-battlefield-autonomy/

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Why do Systems Fail / Succeed?

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Teynampet - Saidapet Steel Flyover, Chennai



Source: https://timesofindia.indiatimes.com/city/chennai/work-picks-up-on-rs-621-crore-anna-salai-flyover-in-chennai-as-spans-arrive/photostory/125536997.cms

"India's first elevated flyover built directly above operational underground Metro Rail tunnels!!..."



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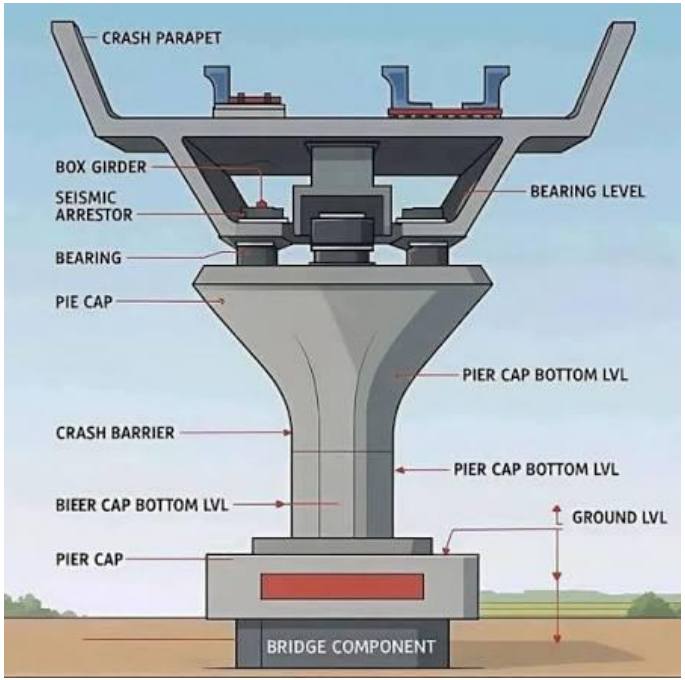
An Engineering Feat?

Pile Cap Design



AI Generated Image

Pre-Fabricated Steel Components



AI Generated Image

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Design flaw leads half-built Chennai flyover back to drawing board

Omjasvin M D / Aug 17, 2026, 23:43 IST

Select TOI as 8 Comments Share AA



The 3.2 km Anna Salai flyover, being built at a cost of Rs 621 crore, is progressing at snail's pace with girders too not being placed between...

The 3.2km Teynampet-Saidapet flyover on Anna Salai will miss its Sept 2026 deadline as the state highways department has run into a major design snag midway.

The ₹621-crore flyover now needs a "scope change" as the department has realised that it cannot proceed with the original pillar design on the narrow Nandanam-Teynampet stretch. The realisation dawned on the officials only after more than 80 pillars, girders and 50 spans were constructed or placed on the Saidapet-Nandanam stretch. The revised deadline is now Feb 2027 — but this too looks uncertain.

The project began in Jan 2024. Earlier, the highways department justified the high project cost by citing its high-tech prefabricated steel structures, which are being brought directly to the site from other states. The structures can be mounted on Metro station foundations, allowing faster construction than conventional concrete flyovers.

But midway through the work, officials found that the DPR did not accurately capture the alignment of the existing Metro foundations on Nandanam-Teynampet stretch. Unlike other sections, the central foundations here are not in a straight line.

"The central foundations of the Metro were not straight from Nandanam to Teynampet.

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What went wrong?



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- The original design assumed that existing Metro infrastructure could be used to support central pillars.
- During construction, engineers found that the Metro foundations in this section did not follow the straight-line alignment assumed in the design.
- Metro Stations in the alignment had box type underground concrete structures; can’t bear the load form central pillar.

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Boeing 737 MAX tragedies



Source: https://en.wikipedia.org/wiki/Lion_Air_Flight_610



Source: https://www.cntraveler.com/stories/2013-04-15/lion-air-plane-crash-in-bali-asian-budget-airlines-041513



Source: https://corporate.ethiopianairlines.com/Press-release-open-page/ethiopian-airlines-commits-to-order-11-boeing-737-max-airplanes



Source: https://www.bloomberg.com/news/articles/2022-12-28/us-aviation-bodies-dispute-parts-of-ethiopian-max-crash-report

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Boeing 737 MAX Crashes: What Happened?

A combination of an erroneous sensor input, flight-control system design, multiple cockpit alerts and high pilot workload led to loss of control in both accidents.

1 AOA sensor gives erroneous reading



2 MCAS activates

MCAS uses AOA input from a single sensor (in the original design).

Flight control computer

MCAS logic

Erroneous high AOA triggers MCAS

Command nose-down stabilizer trim

3 Stabilizer moves nose down



MCAS commands repeated nose-down trim inputs (about 2.5° per activation in the original design).

4 Aircraft pitches down



5 Pilots respond, but MCAS re-activates

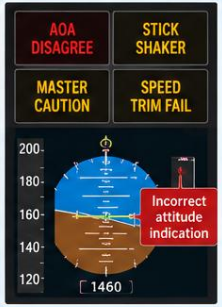
Pilots try to counter the pitch down using control column and electric trim switches.



When pilots use electric trim to counter the trim, MCAS can re-activate and command nose-down trim again as long as the erroneous high AOA input persists.

6 Multiple alerts increase pilot workload

The crew experiences multiple warnings and indications at the same time, making it more difficult to diagnose the problem.



Typical sequence of events (simplified)



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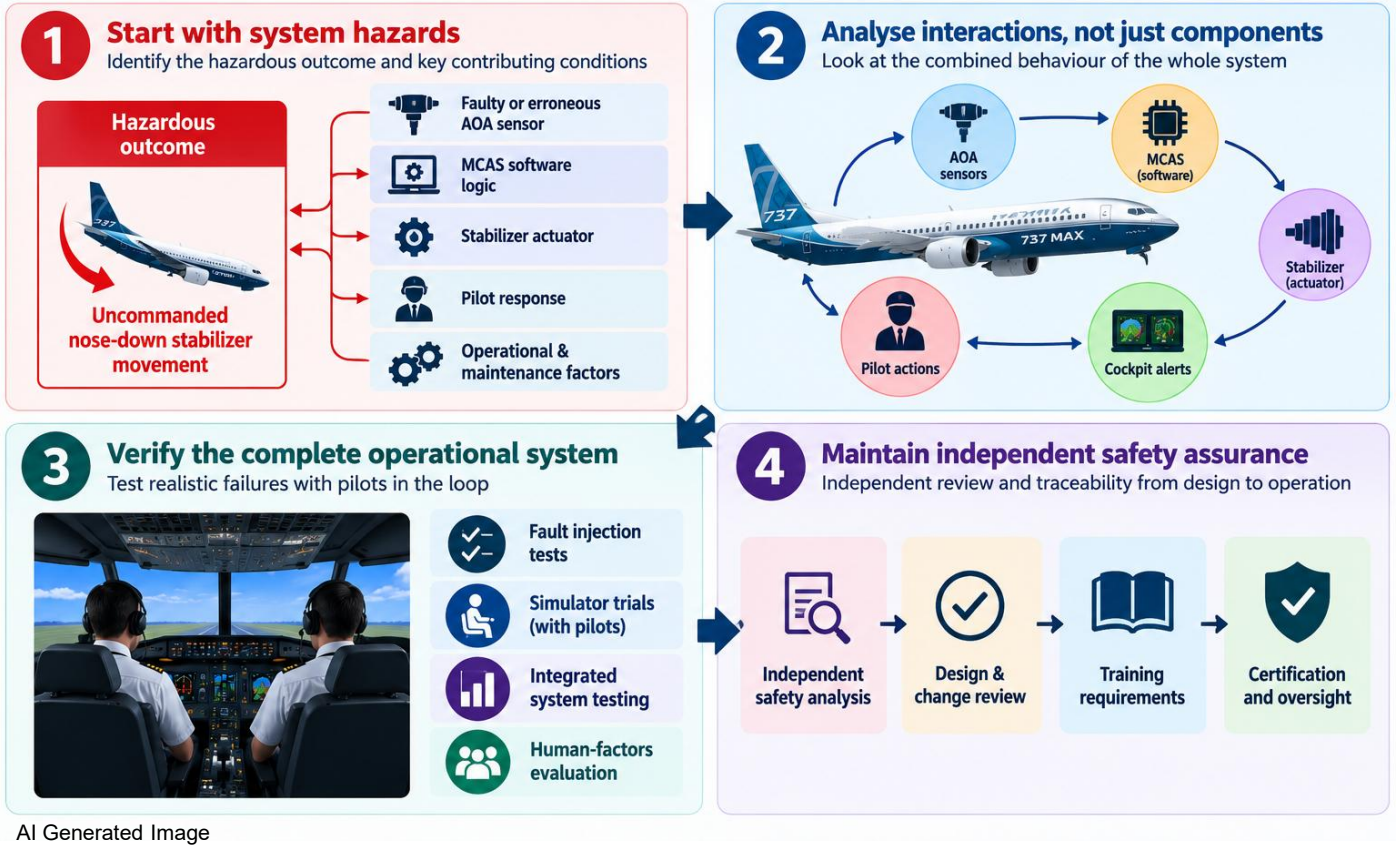
What caused it?

Trigger	Erroneous AoA sensor input
Flight-control mechanism	MCAS activation and repeated nose-down trim commands
Operational challenge	Pilots faced difficult aircraft-control conditions alongside multiple alerts and indications
Maintenance contribution	A miscalibrated replacement sensor contributed to the Lion Air accident sequence
Systemic weaknesses	Safety-assessment assumptions, design safeguards, training and documentation, and certification oversight

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What could have prevented it?

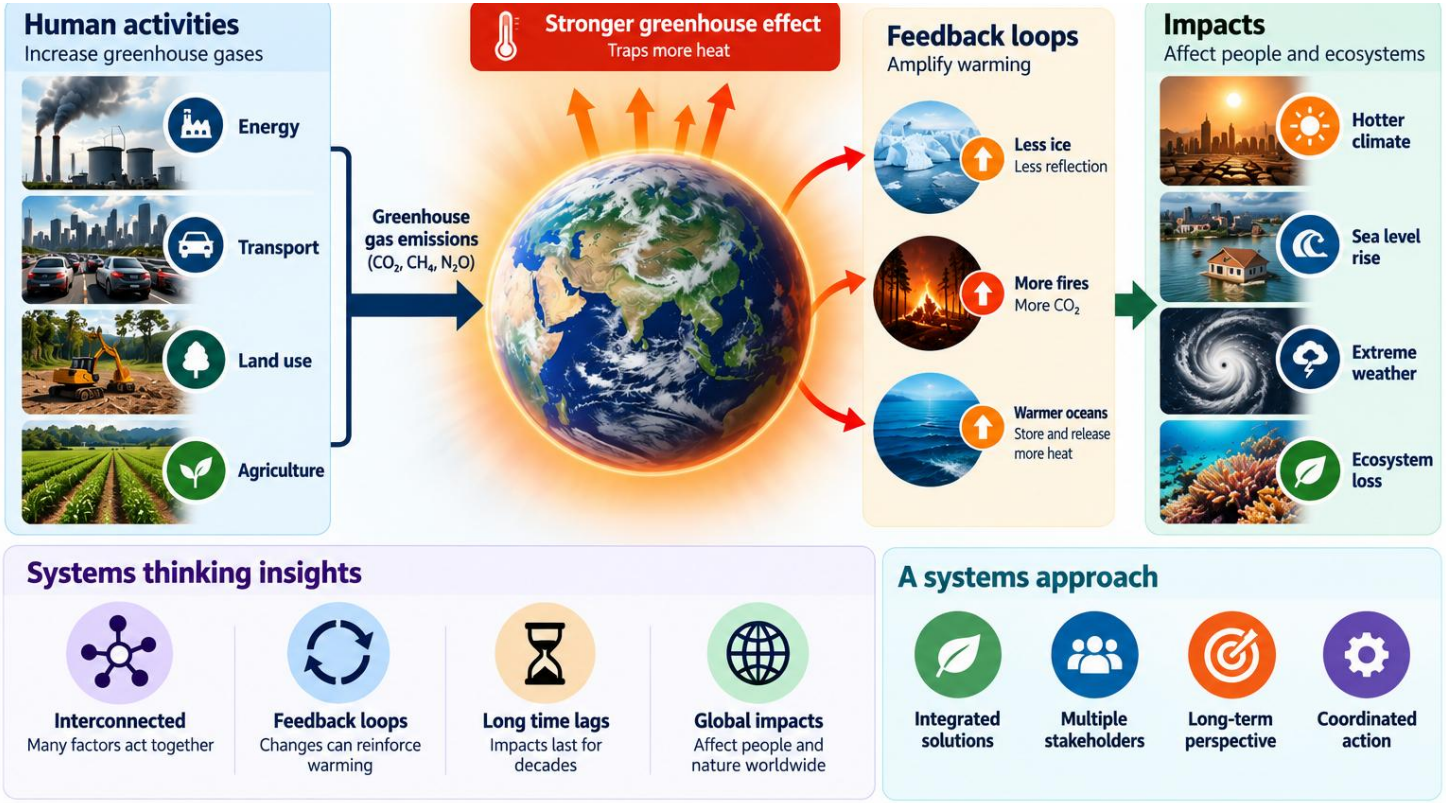


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Global warming – a systems problem

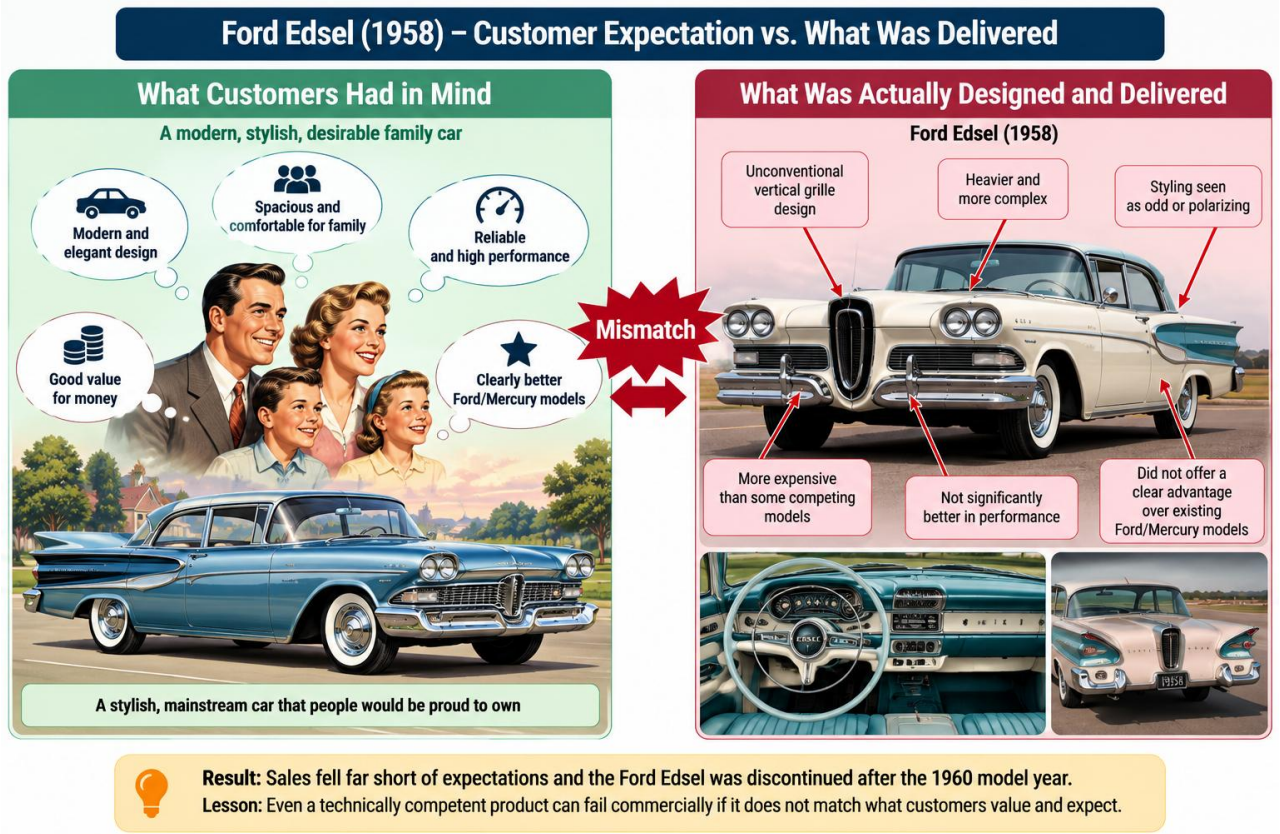


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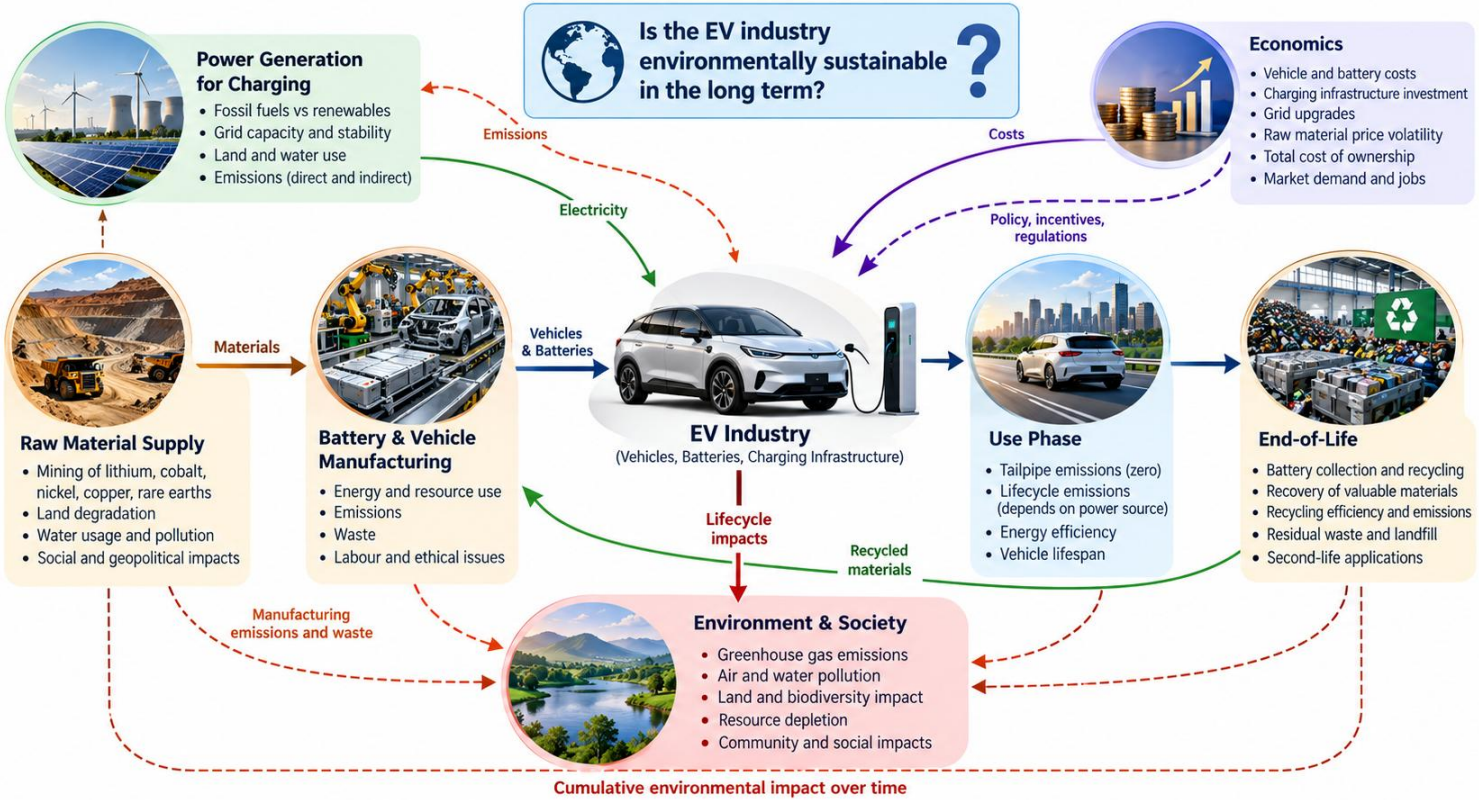
Market Tragedies



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Sustainable Solution?



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“Most failures occur not because nobody understood how a system components work. They occur because nobody adequately understood how the components, people, processes, and environment behaved together.”

“Yet we act as if simple cause and effect is at work. We push to find the one simple reason things have gone wrong. We look for the one action, or the one person, that created this mess. As soon as we find someone to blame, we act as if we’ve solved the problem.”

– Margaret J. Wheatley

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What about successful systems?

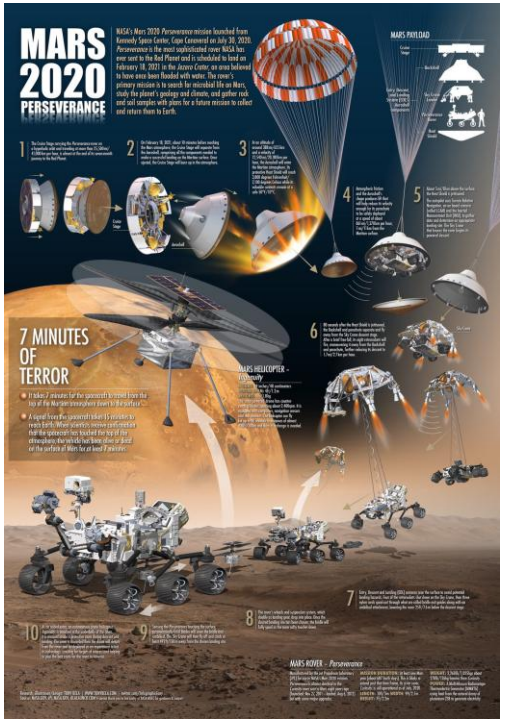
“First Time Right” Paradigm



Source: https://www.nasa.gov/mission/apollo-11/



Source: https://www.instagram.com/reel/Da0k54pRwDt/



Source: https://www.humanmars.net/2021/02/nasa-mars-2020-perseverance-rover.html

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What made them Successful?

Rigorous, Uncompromising

Systems Engineering

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Systems Engineering

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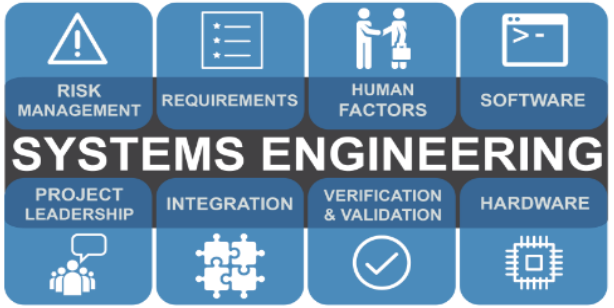
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System Engineering - The Saviour

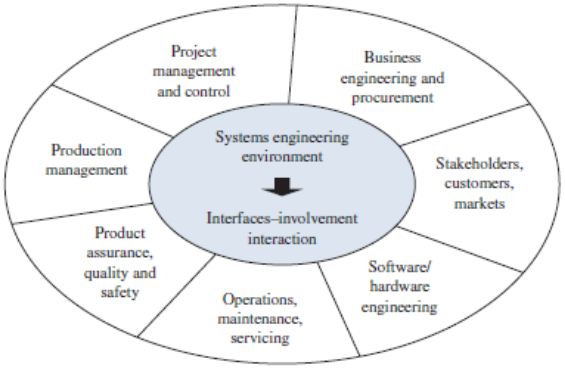
- **Systems Engineering (SE)** is an interdisciplinary approach based on **Systems Thinking** to enable the realization of successful systems.

- It focuses on defining **customer needs** and **required functionality** early in the **development cycle**, documenting **requirements**, and then proceeding with **design synthesis** and **system validation** while considering the complete problem: **operations, cost and schedule, performance, training and support, test, manufacturing, and disposal**.

- SE considers both the **business and the technical needs** of all **customers** with the goal of providing **a quality product** that **meets the user needs**. (INCOSE)



Source: INCOSE Systems Engineering Handbook V5



Source: INCOSE Systems Engineering Handbook V5

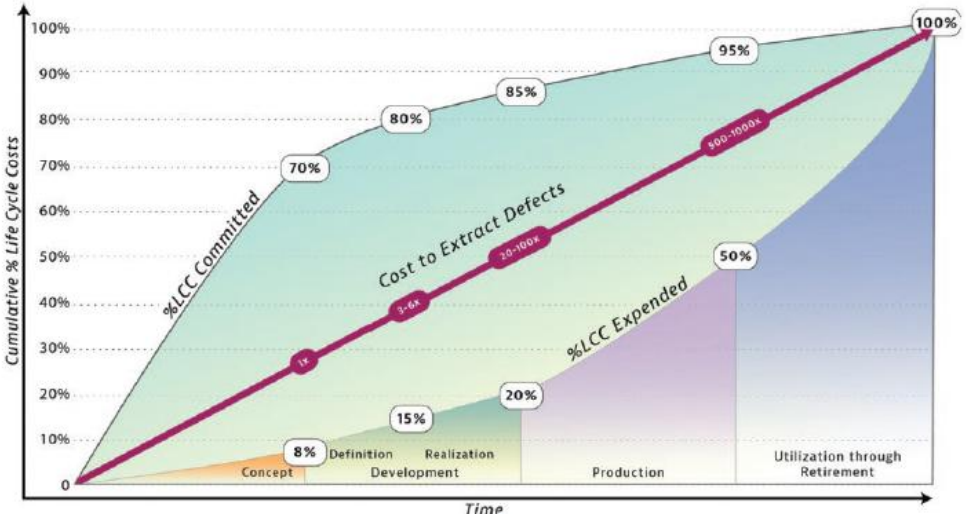
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System Engineering

Factors driving SE need

- System complexity and criticality
- Rapid change in technology
- Continuous change in requirements
- Risk reduction at early stages
- Tight cost and schedule constraints
- Competitive product development
- Short development cycles

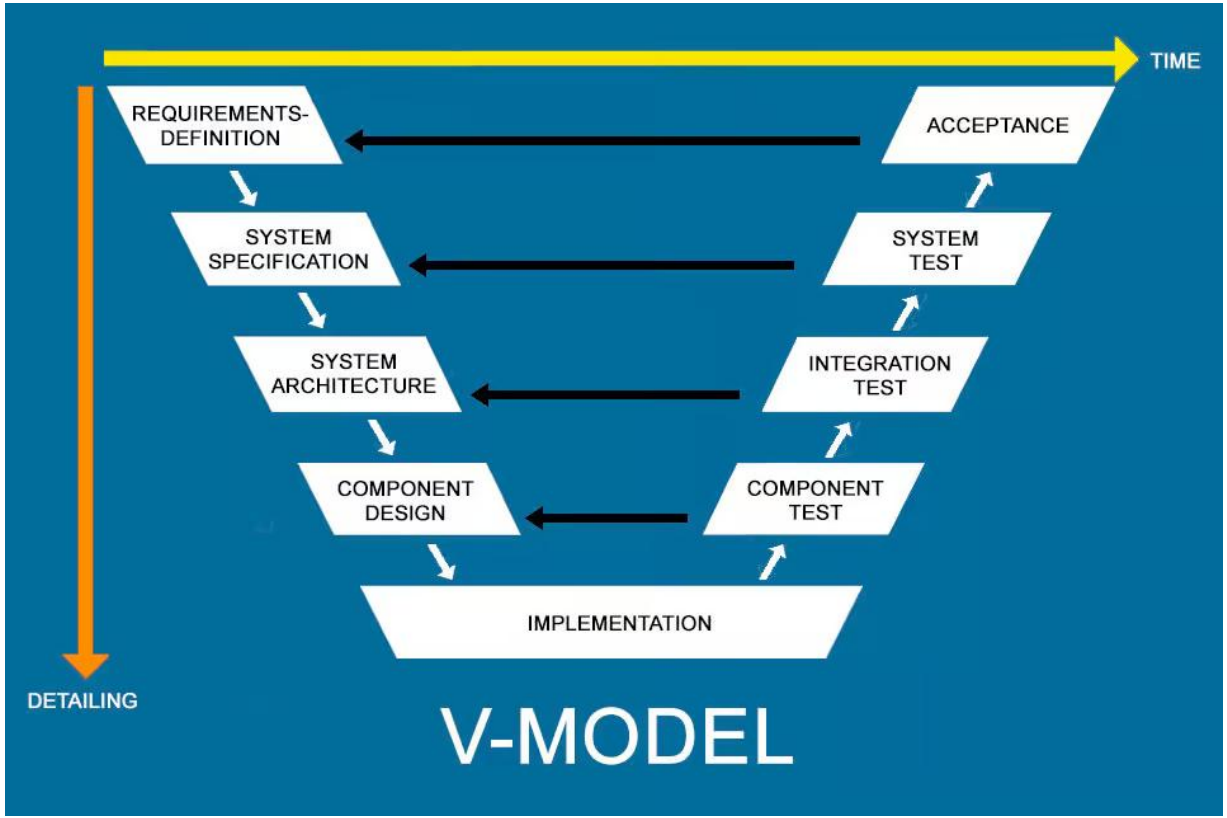


Source: INCOSE Systems Engineering Handbook V5

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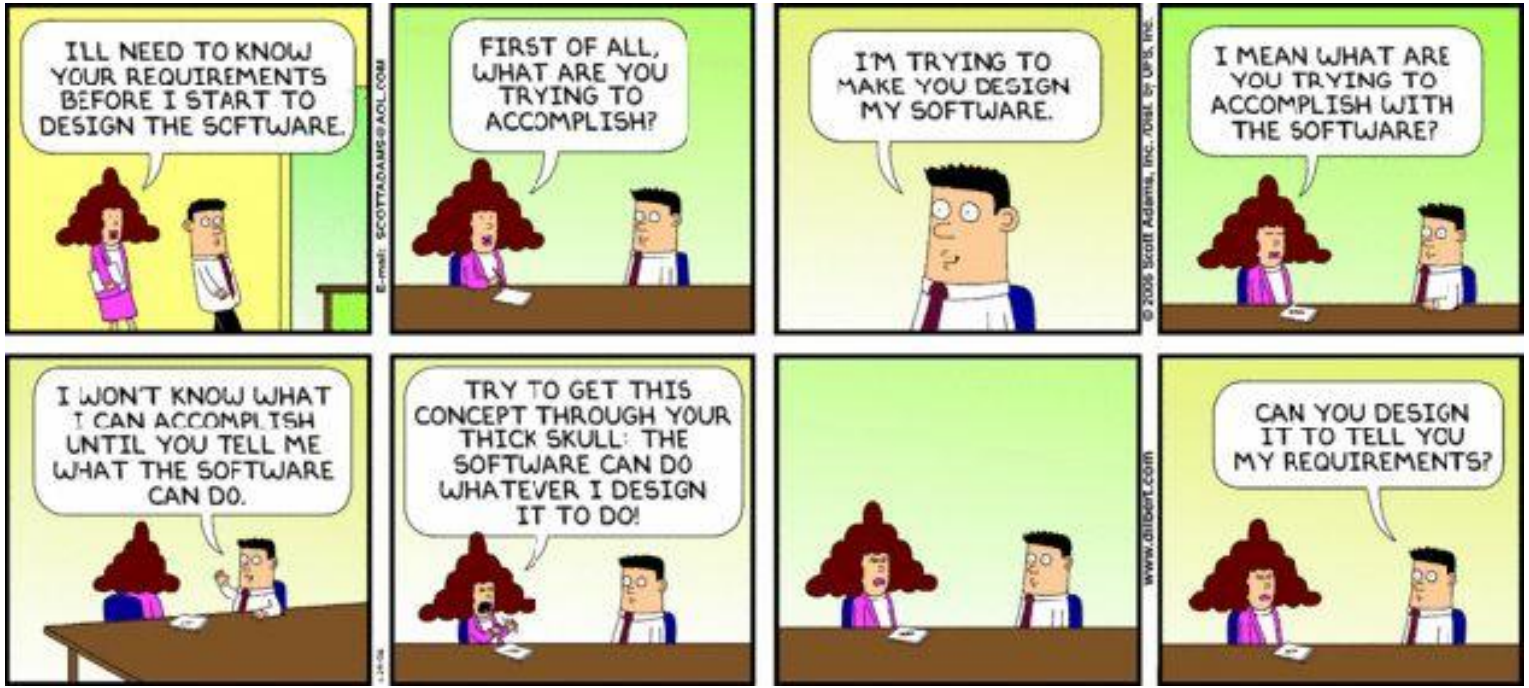
System Engineering



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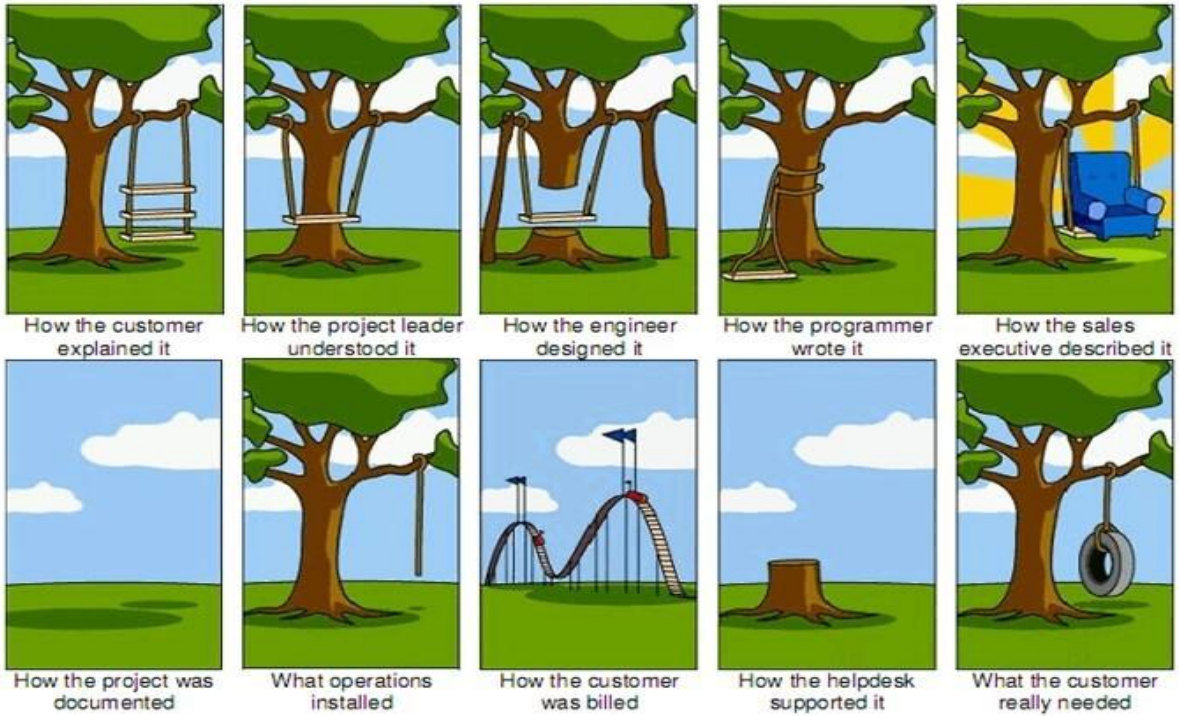
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What system to build?



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System Quality? That is a Quality inspector's job..



Source: https://engineering.cerner.com/blog/learn-what-the-rules-dont-cover/

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System Requirements

The Single Source of Truth that drives successful product development

- An **agreement between the stakeholders** on what needs to be built and/or delivered.
- A **formal record that serves as the baseline** for system development or service delivery.
- Generally, answers the question of **“what”** needs to be built **rather than “how”.**
- **Primary basis** for performing the **verification and validation** of the system.
- Output of every phase of system development should be **traced to the requirements**.

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System Requirements

Types of Requirements

Safety Requirements
Ensures product safety and compliance with regulations.

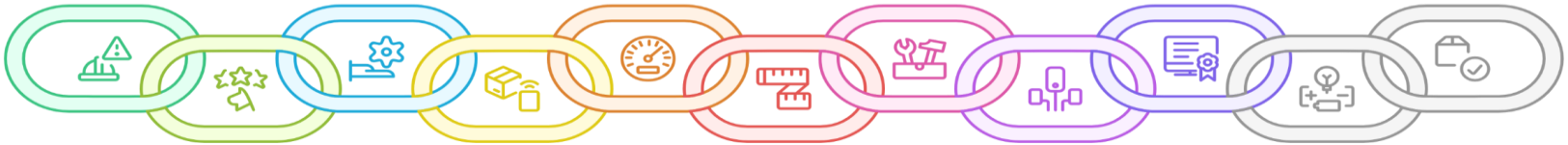
Functional Requirements
Defines the product's core functions and features.

Performance Requirements
Sets standards for product performance and efficiency.

Maintainability Requirements
Ensures ease of maintenance and repair.

Certification Requirements
Ensures compliance with industry standards and certifications.

Product Qualification Requirements
Ensures product meets quality and performance standards.



Customer Requirements
Addresses customer needs and expectations.

Operational Requirements
Specifies conditions for product operation and usage.

Physical and Installation Requirements
Outlines physical dimensions and installation guidelines.

Interface Requirements
Defines interactions with other systems and users.

Derived Requirements
Requirements that emerge from the design process.

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System Requirements

Characteristics of good requirements

THE 9 SUPER POWERS



Our hero discovers the foundation stones of requirement excellence

FOUNDATION POWERS

- NECESSARY - Every requirement must serve a purpose
- APPROPRIATE - Must fit the context and stakeholder needs
- CORRECT - Accurately reflects the true business need



The crystal clarity gems that eliminate confusion and misunderstanding

CLARITY GEMS

- UNAMBIGUOUS - Clear meaning with no room for interpretation
- COMPLETE - Contains all necessary information and details
- SINGULAR - Addresses one specific requirement at a time



The implementation shields that ensure requirements can become reality

REALITY SHIELDS

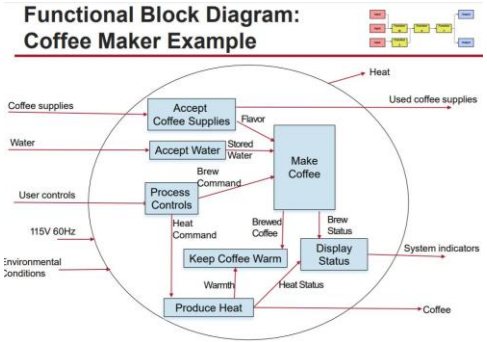
- FEASIBLE - Can be implemented within constraints and resources
- VERIFIABLE - Can be tested and proven to work
- CONSISTENT - Aligns with other requirements without conflicts

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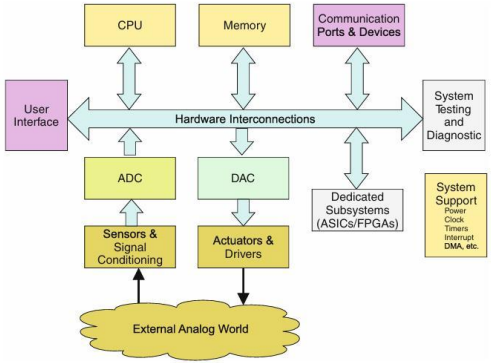
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System Architecture Views

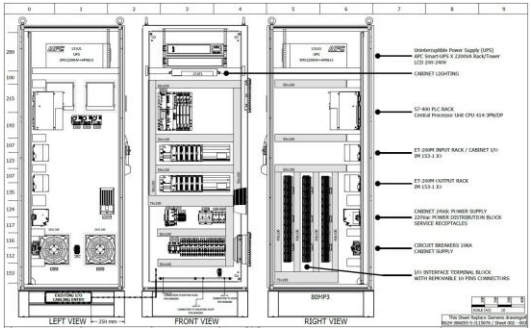
Functional - What the system does?



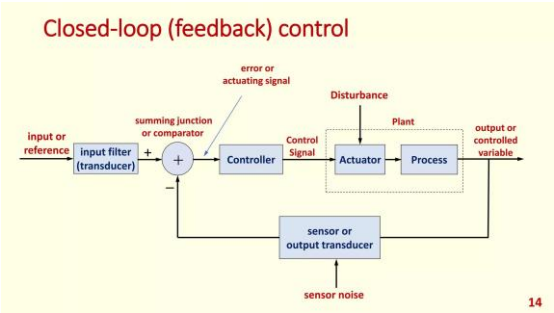
Physical - How is it structured?



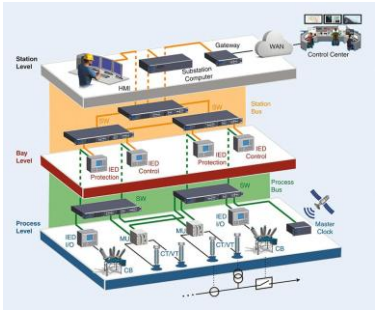
Installation - How is it deployed?



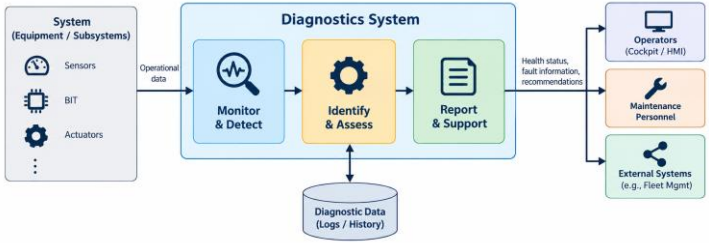
Control - How is it controlled?



Communication - How does it communicate?



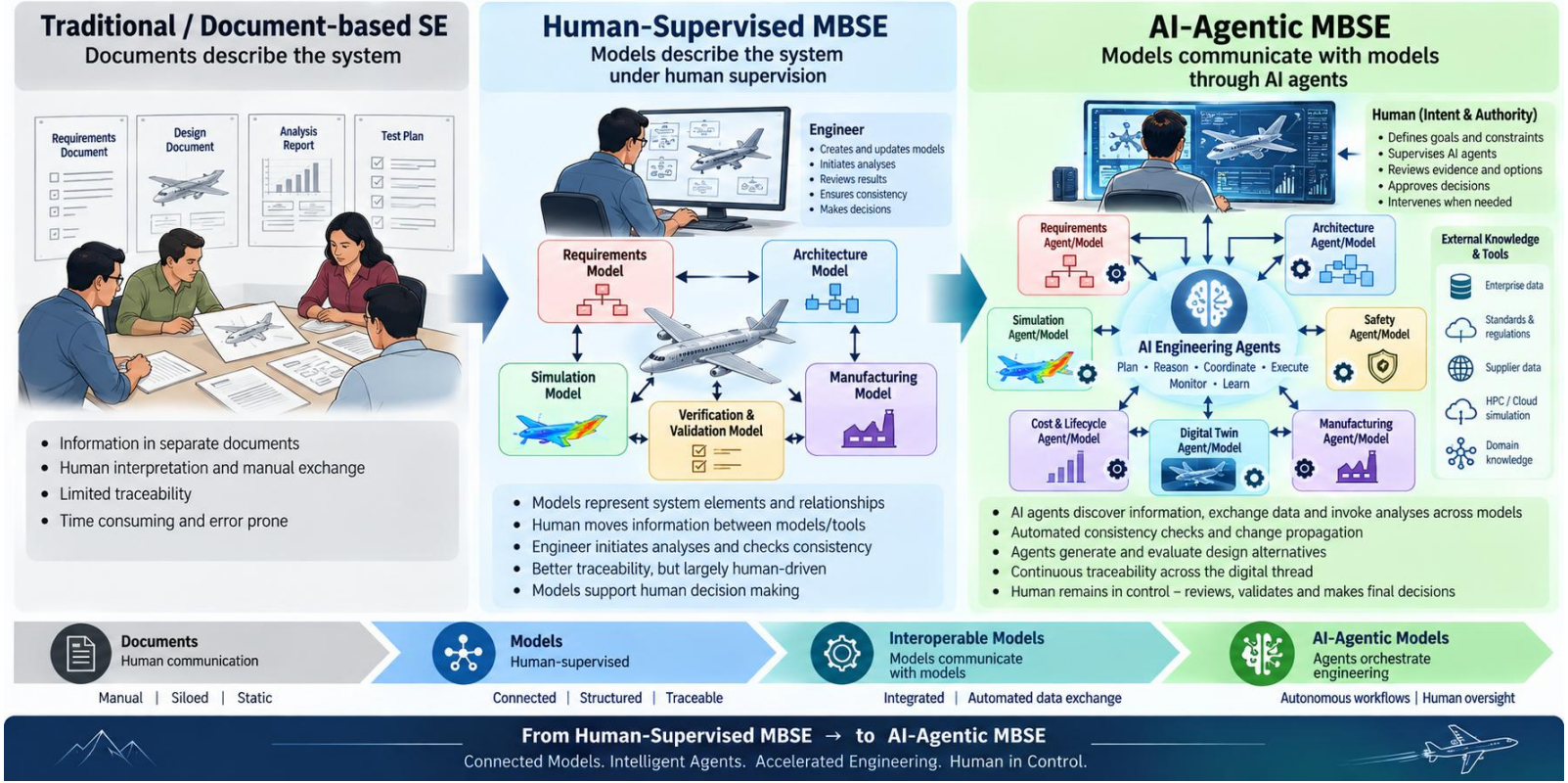
Health management - How is its health managed?



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Model Based System Engineering

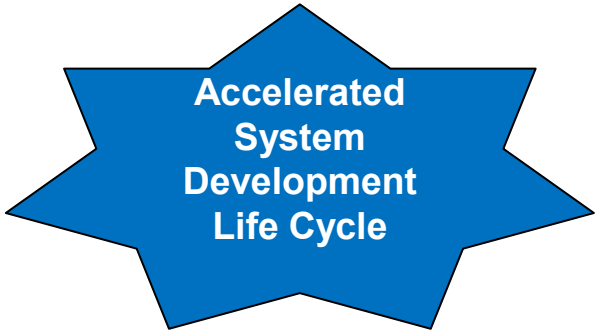


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Model Based System Engineering





- Model to model communication
- Automated consistency checks
- Intelligent information discovery
- Automated analysis and simulation
- Generation and Evaluation of alternatives





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“All things appear and disappear because of the concurrence of causes and conditions. Nothing ever exists entirely alone; everything is in relation to everything else.”

– Buddha

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Q & A



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THANK YOU

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